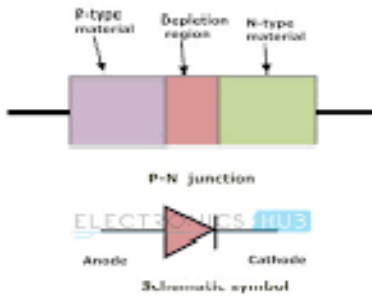
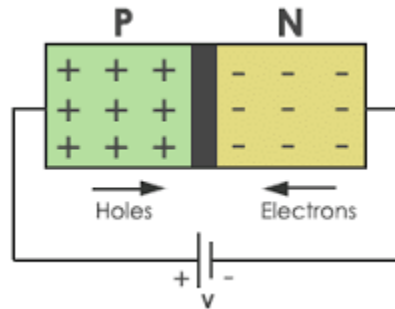


UNIT- I

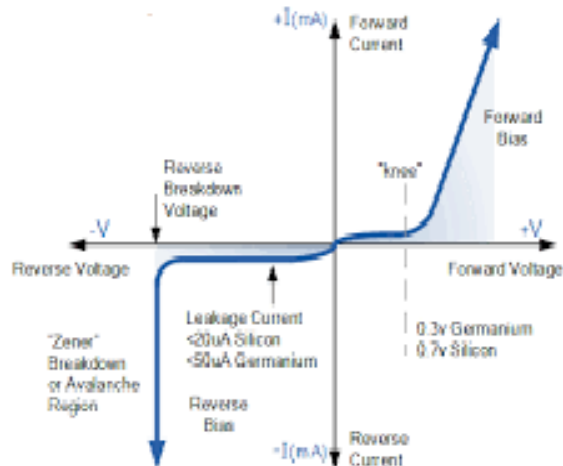
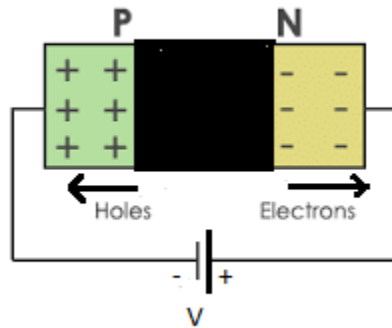
1. PN JUNCTION DIODE:



Forward biasing:

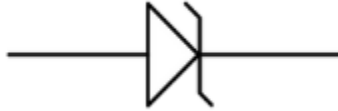


Reverse biasing:

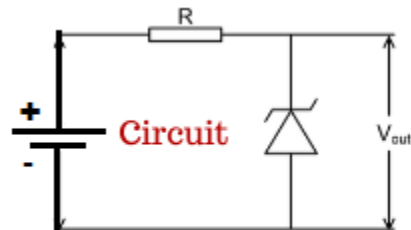


2. ZENER DIODE:

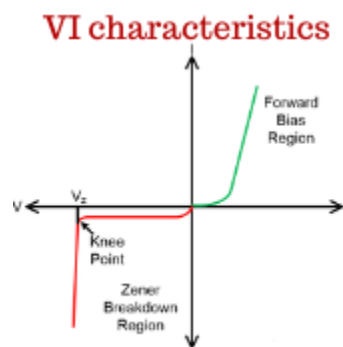
Symbol



Circuit Diagram:

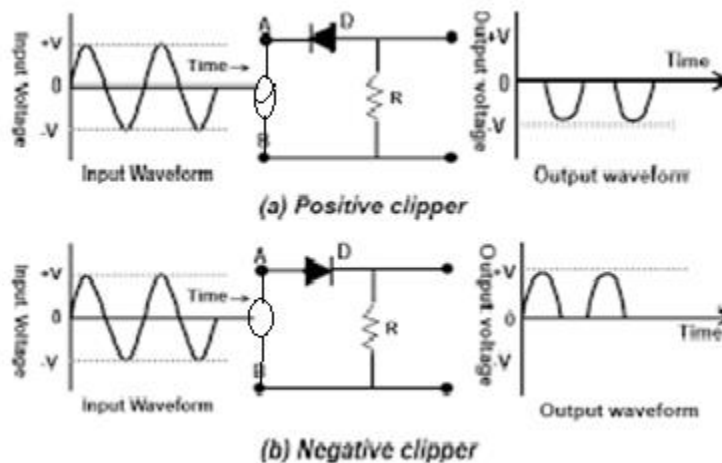


Graph:

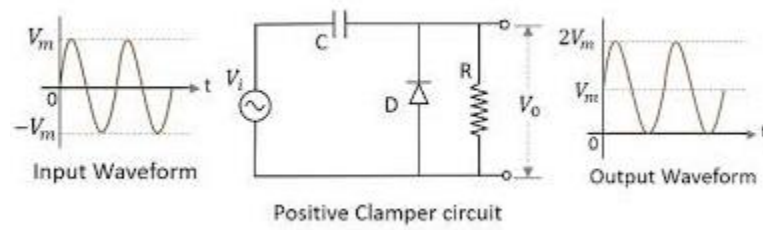


3. CLIPPER AND CLAMPER:

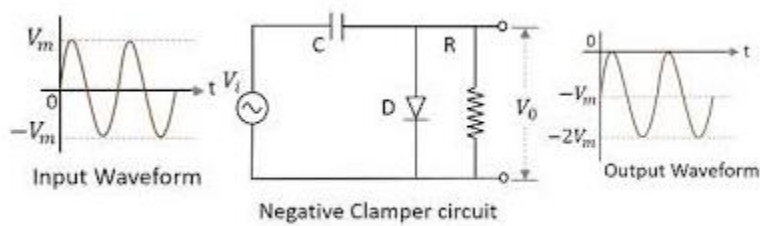
Clipper Circuit Diagram



Clamper Circuit Diagram



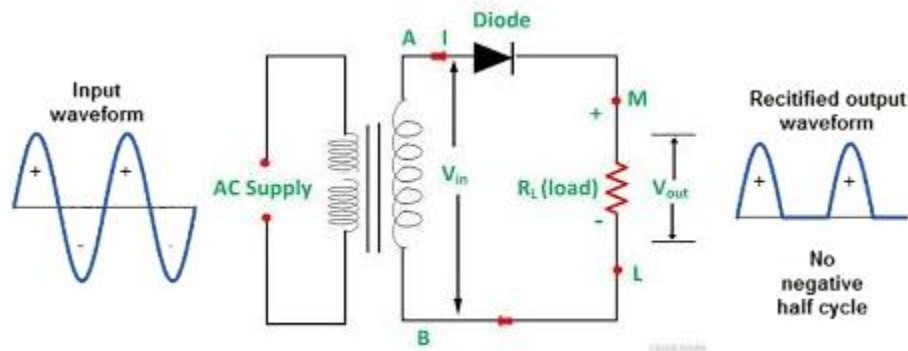
(a) Positive clamper



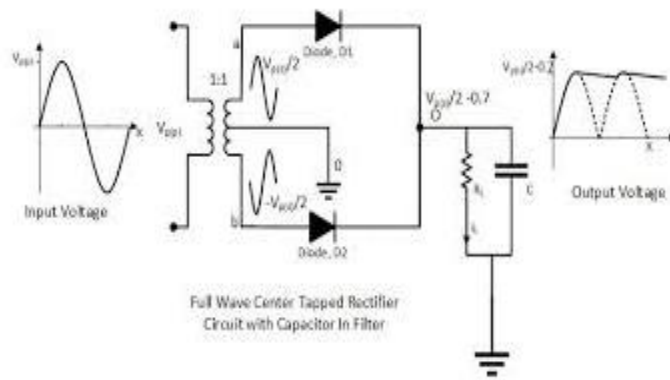
(b) Negative Clamper

4. RECTIFIER:

Halfwave rectifier:

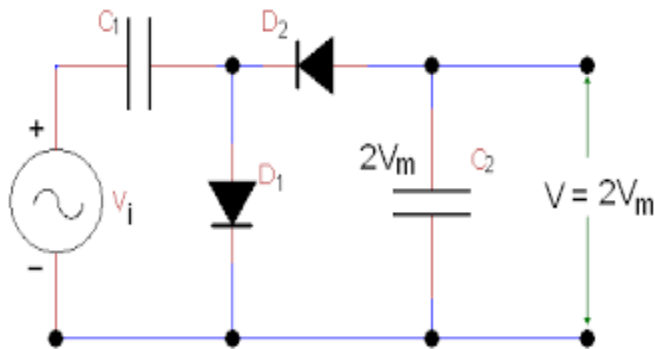


Fullwave Rectifier With Filter:

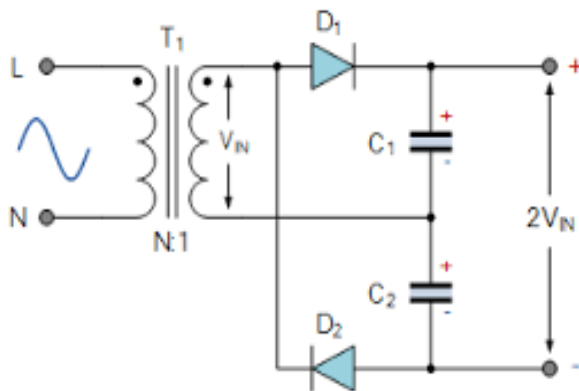


5. VOLTAGE MULTIPLER:

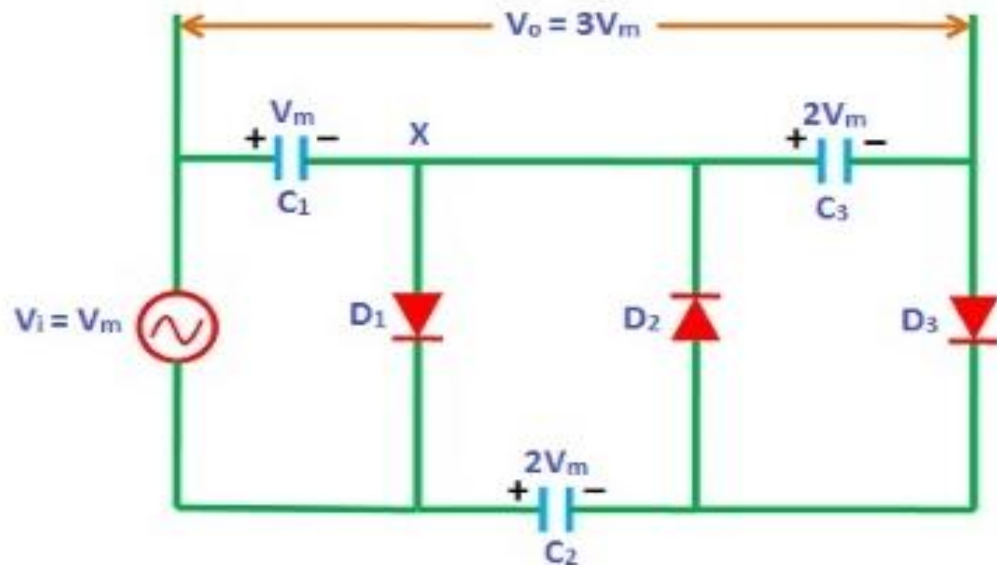
Half wave voltage doubler:



Full wave voltage doubler:

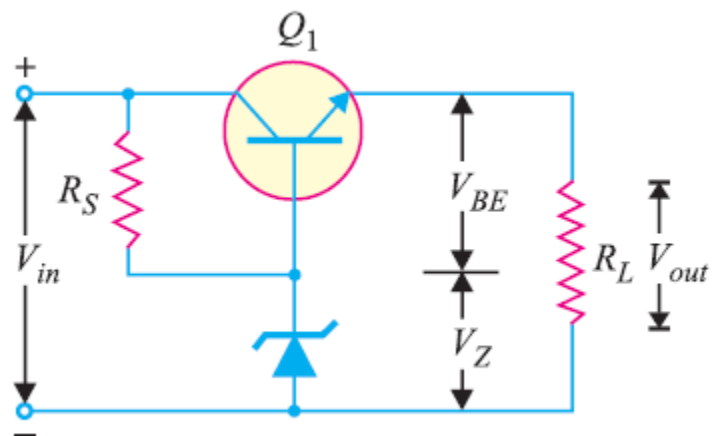


Voltage Tribler:

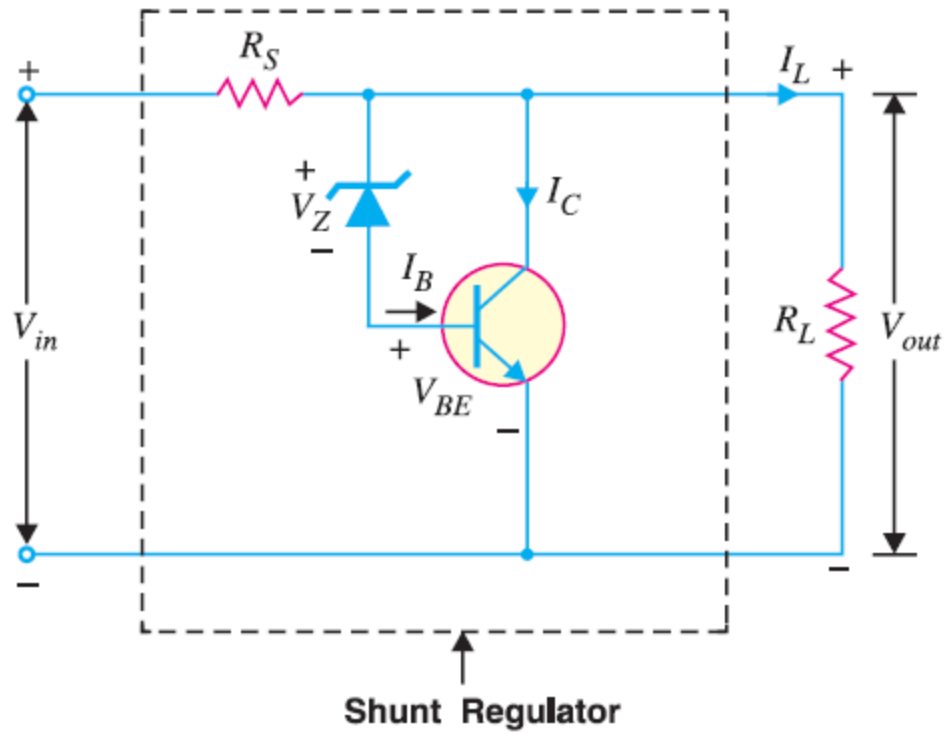


6. VOLTAGE REGULATOR:

Series Voltage Regulator:



Shunt Voltage Regulator:



7. OR and AND GATE USING DIODE:

OR Circuit:

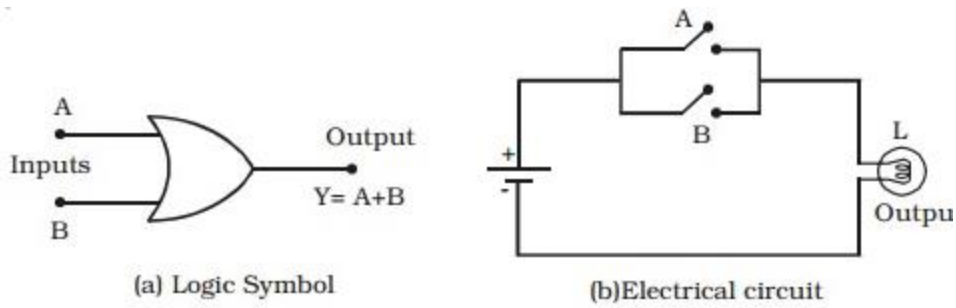


Fig OR gate

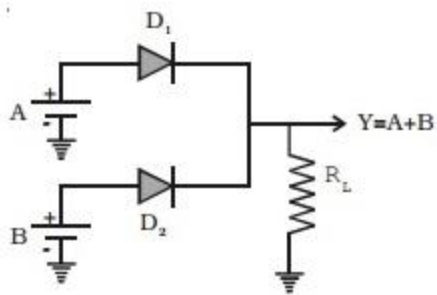


Fig OR gate using diodes

Table Truth table of OR gate

Inputs		Output
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

AND Circuit:

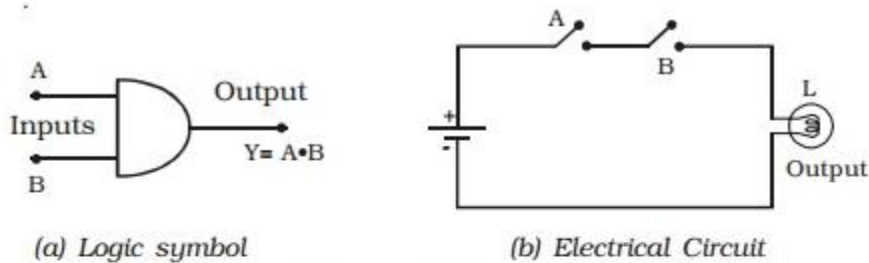


Fig AND gate

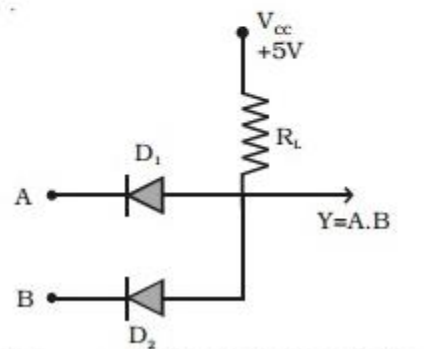


Fig AND gate using diodes

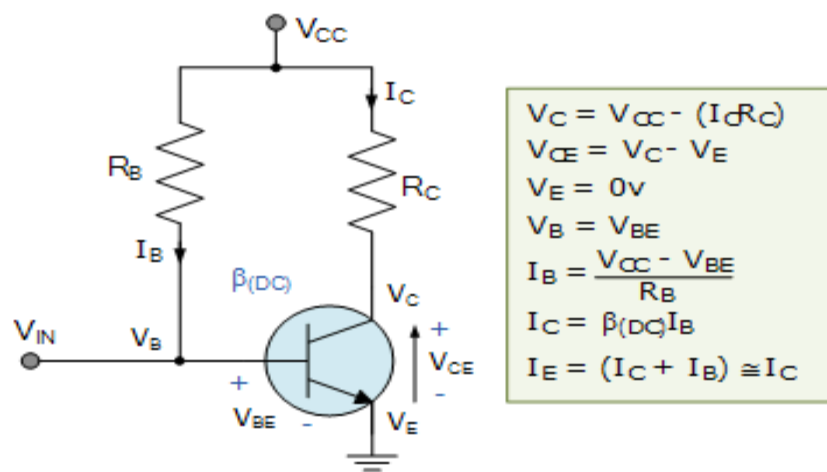
Table Truth table of AND gate

Inputs		Output
A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

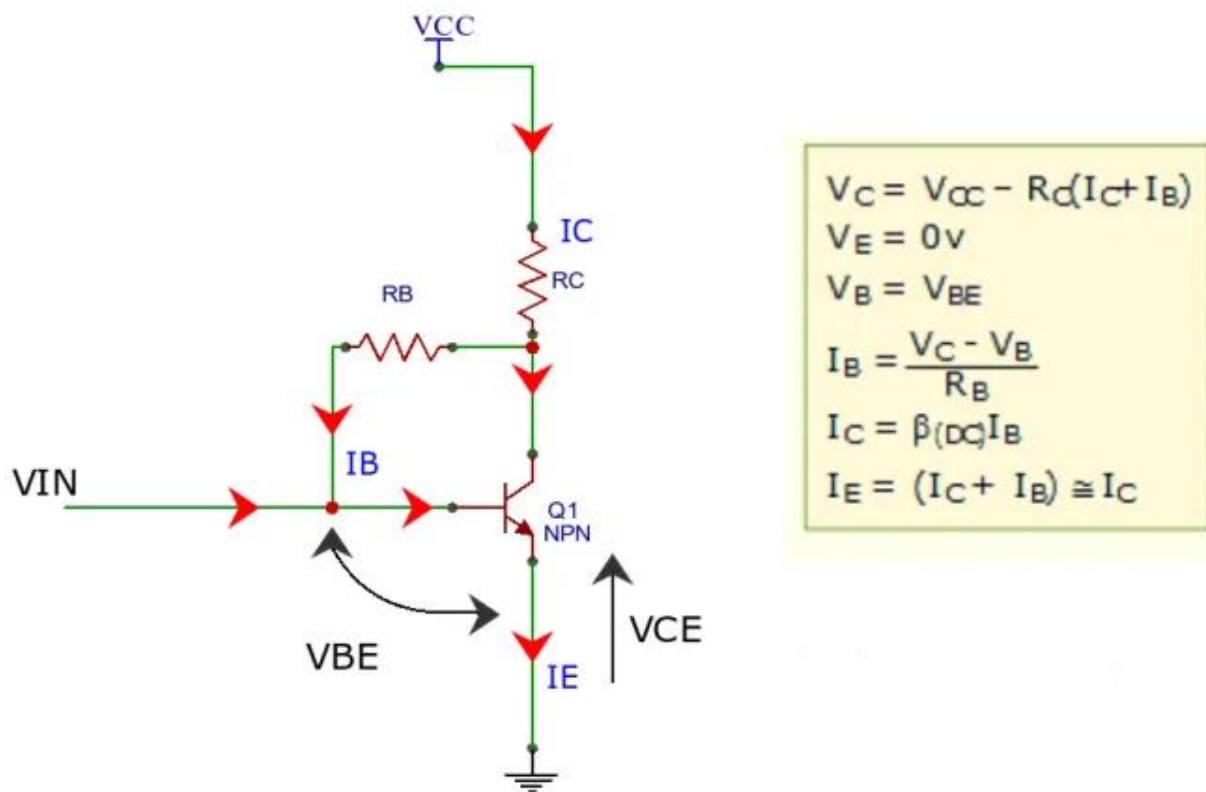
UNIT- II

1. TRANSISTOR BIASING:

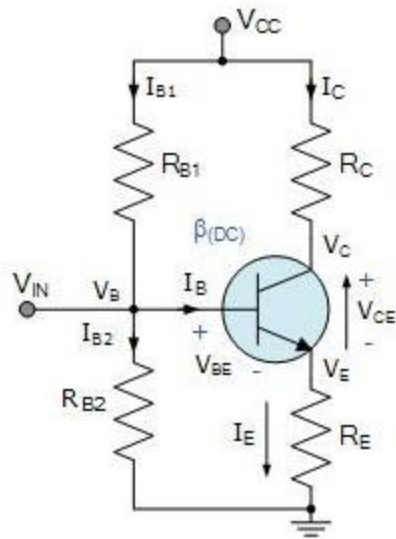
(a) Fixed bias:



(b) Collector to base bias:



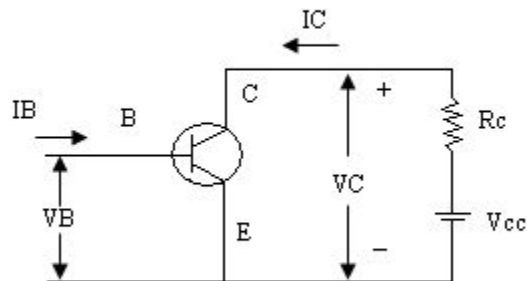
(c) Self or Voltage divider bias:



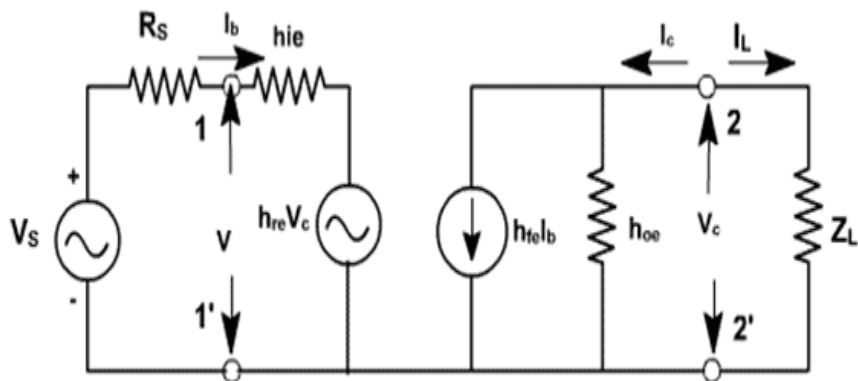
$$\begin{aligned}
 V_C &= V_{CC} - R_C I_C = V_E + V_{CE} \\
 V_E &= I_E R_E = V_B - V_{BE} \\
 V_{CE} &= V_C - V_E = V_{CC} - I_C (R_C + R_E) \\
 V_B &= V_{BE} + V_E = V_{RB2} = \left(\frac{R_{B2}}{R_{B1} + R_{B2}} \right) V_{CC} \\
 I_{B2} &= \frac{V_B}{R_{B2}} \\
 I_{B1} &= I_B + I_{B2} = \frac{V_C - V_B}{R_{B1}} \\
 I_B &= \frac{V_B}{R_B + (1 + \beta) R_E} \\
 I_C &= \beta(DC) I_B \\
 I_E &= I_C + I_B = \frac{V_E}{R_E}
 \end{aligned}$$

2. H- PARAMETER COMMON EMITTER CONFIGURATION:

Equivalent circuit:



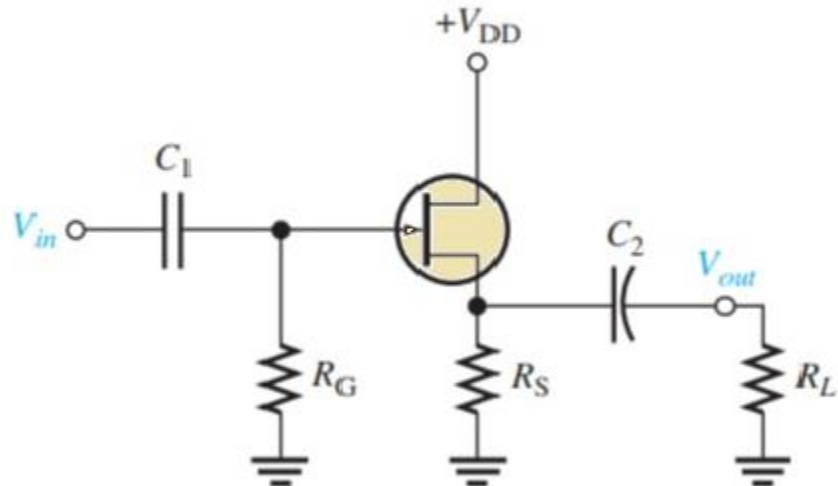
H – parameter:



3. JFET:

(a) Common Drain:

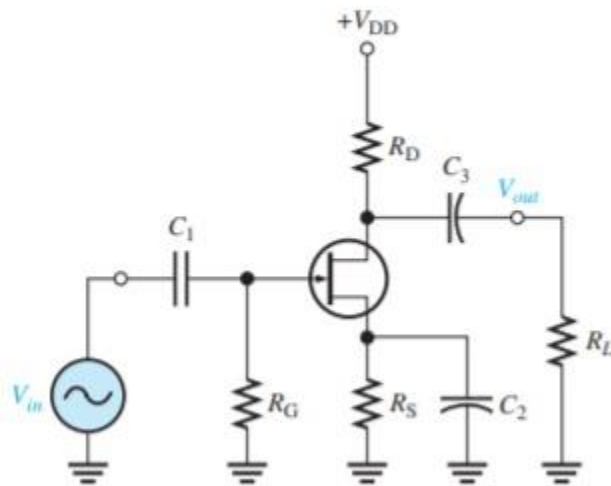
Common-Drain FET Amplifiers



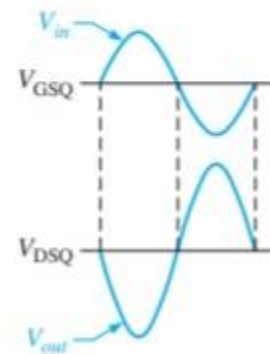
JFET common-drain amplifier (source-follower)

(b) Common Source:

JFET common-source amplifier

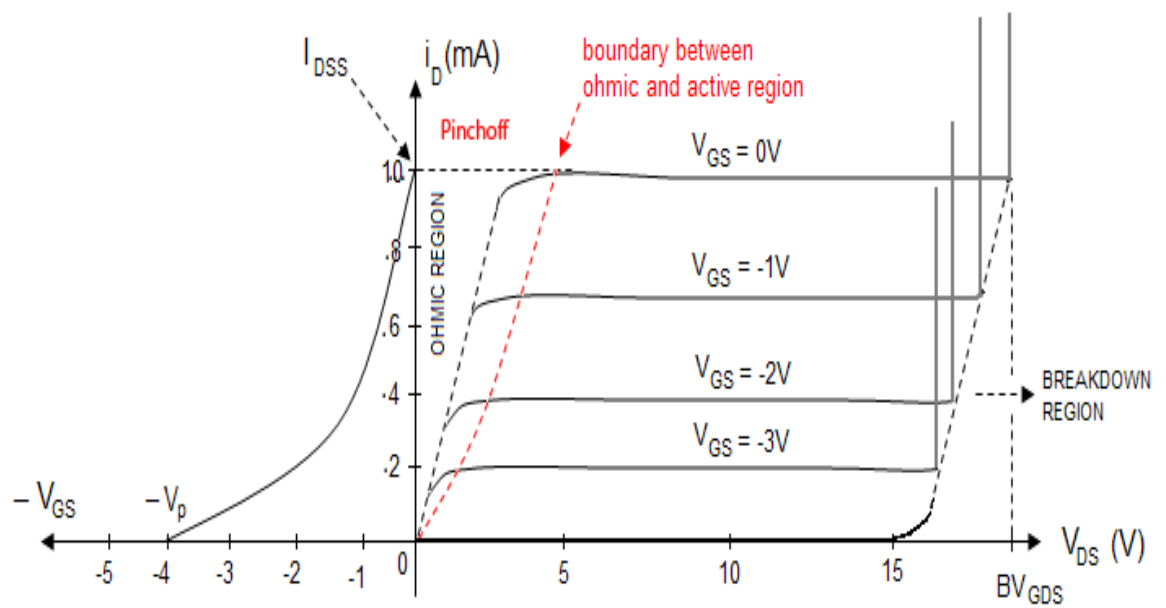
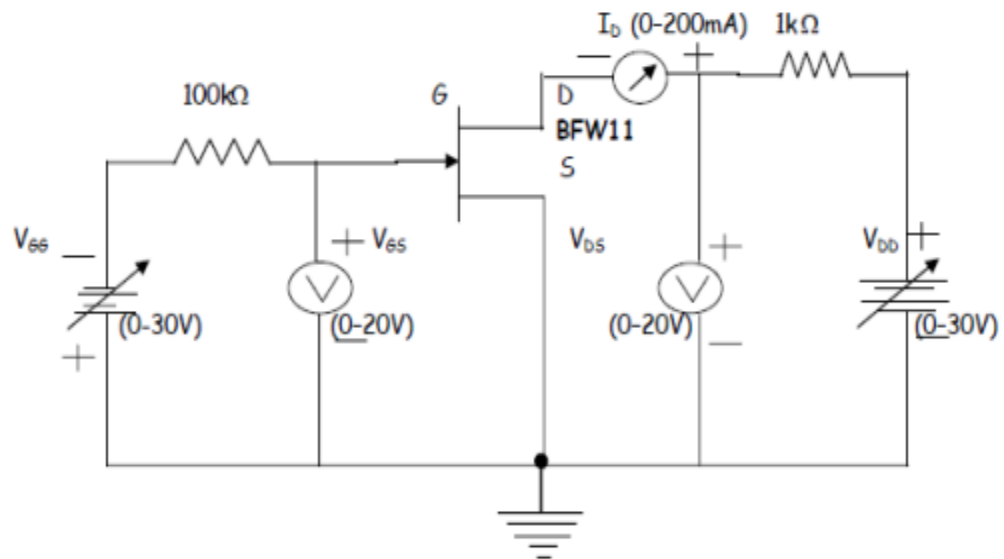


(a) Schematic



(b) Voltage waveform relationship

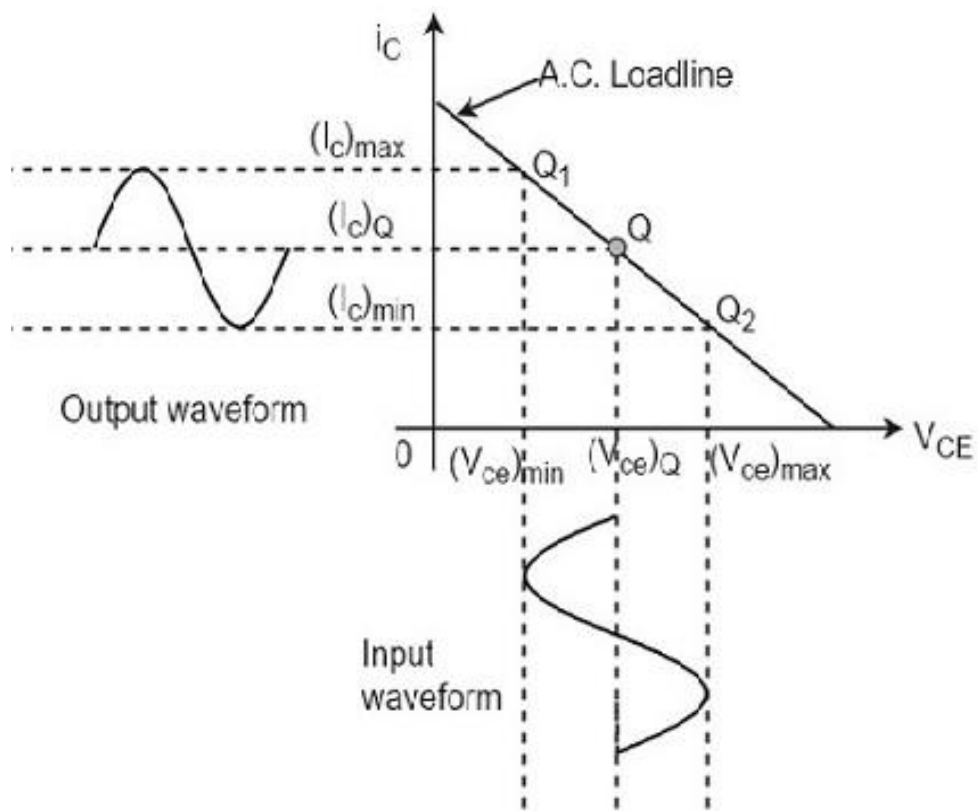
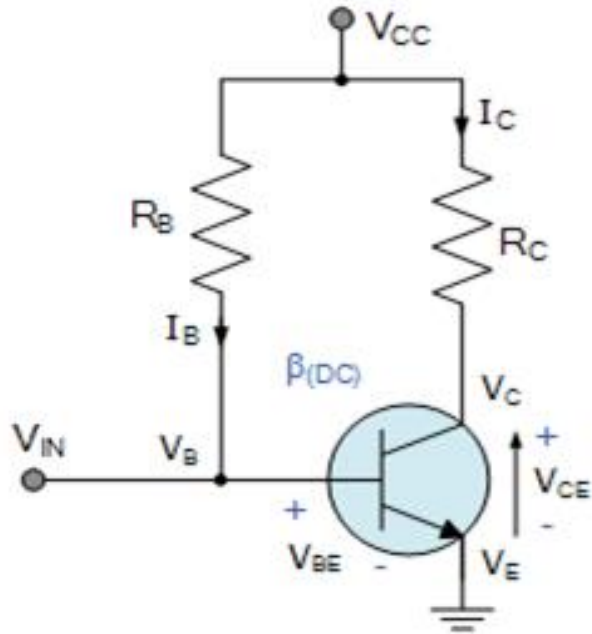
4. JUNCTION FIELD EFFECT TRANSISTOR:



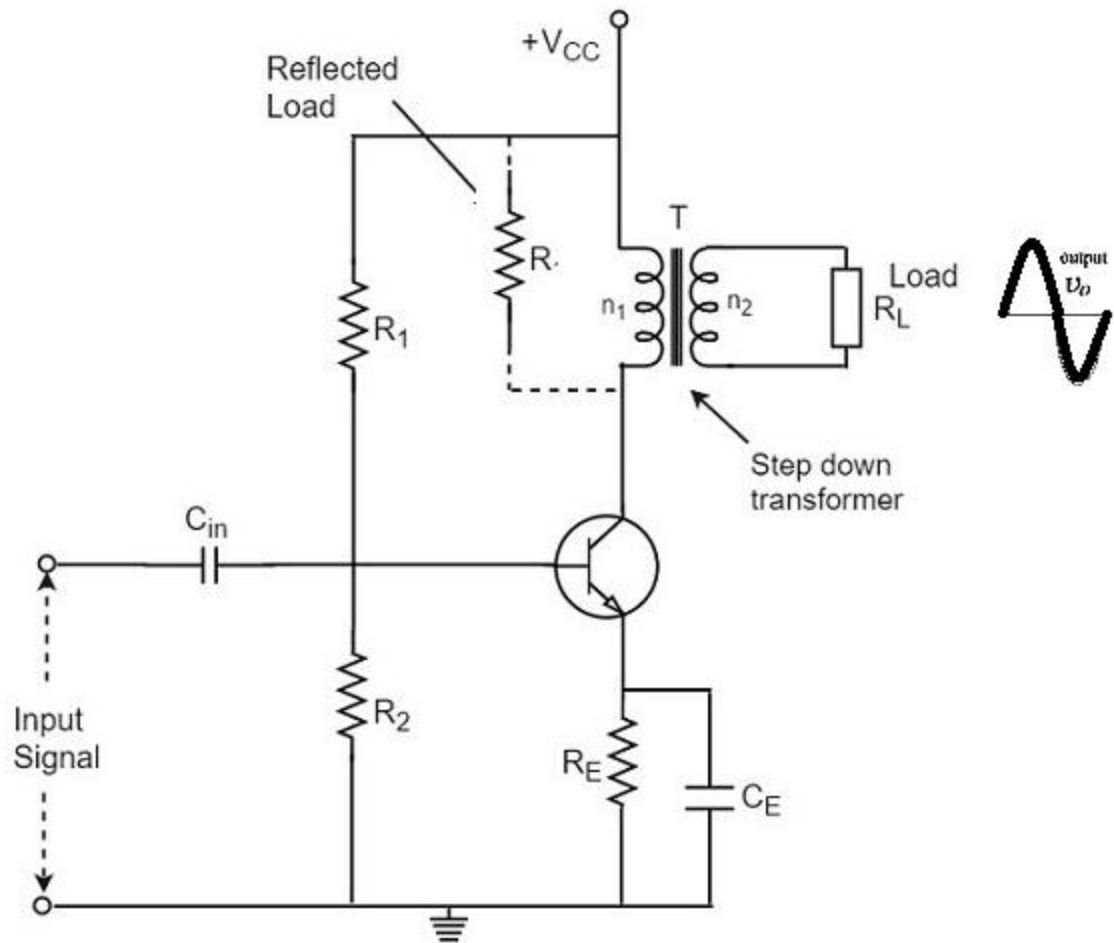
Transfer characteristics

Drain characteristics

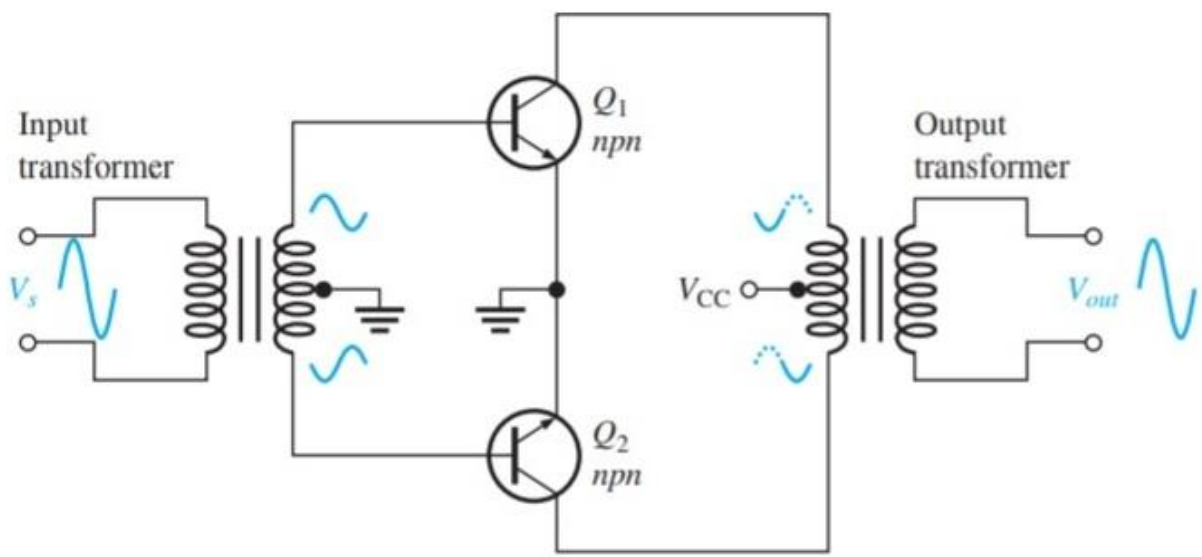
1. CLASS 'A' SERIES FED AMPLIFIER:



2. CLASS 'A' TRANSFORMER COUPLED AMPLIFIER:

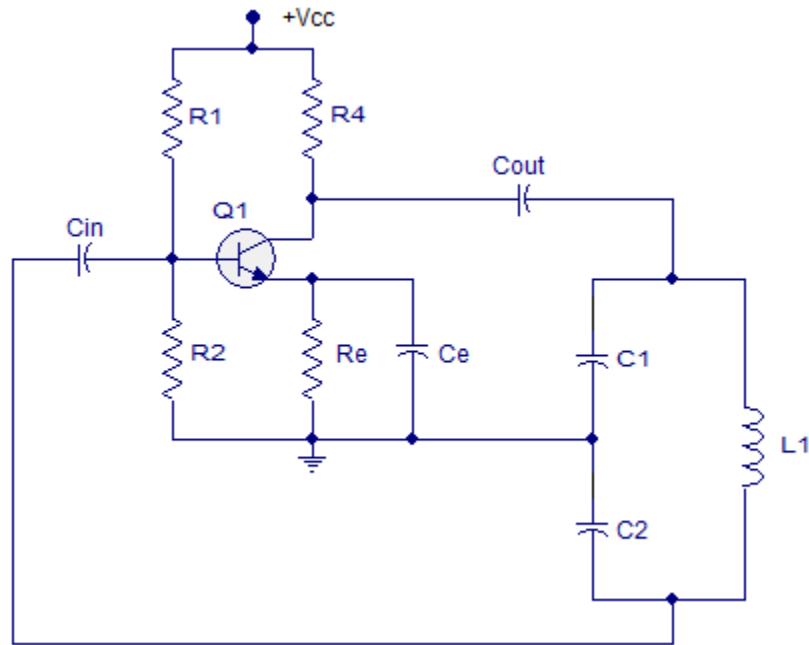


3. CLASS 'B' TRANSFORMER COUPLED PUSH - PULL AMPLIFIER:

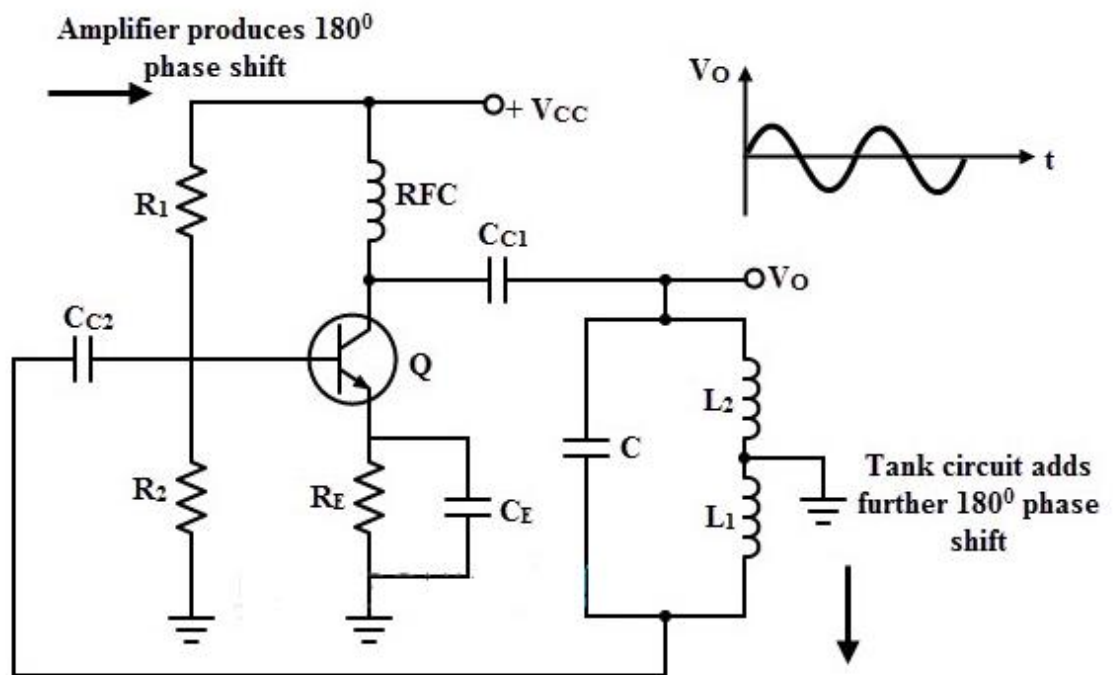


4. OSCILLATOR:

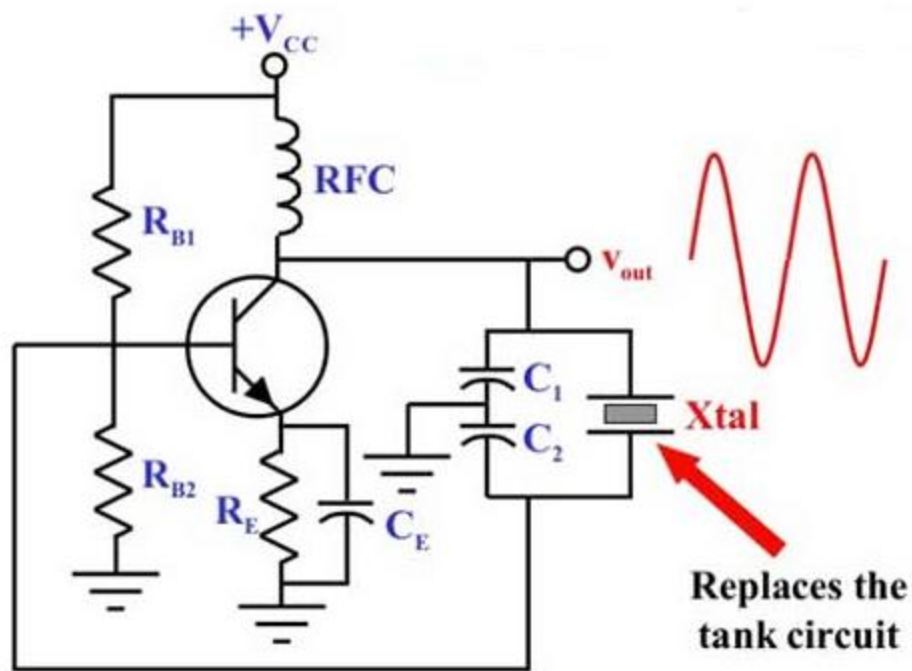
(A) Colpitts Oscillator:



(B) Hartely Oscillator:



(C) Crystal Oscillator:



5. WEIN BRIDGE OSCILLATOR:

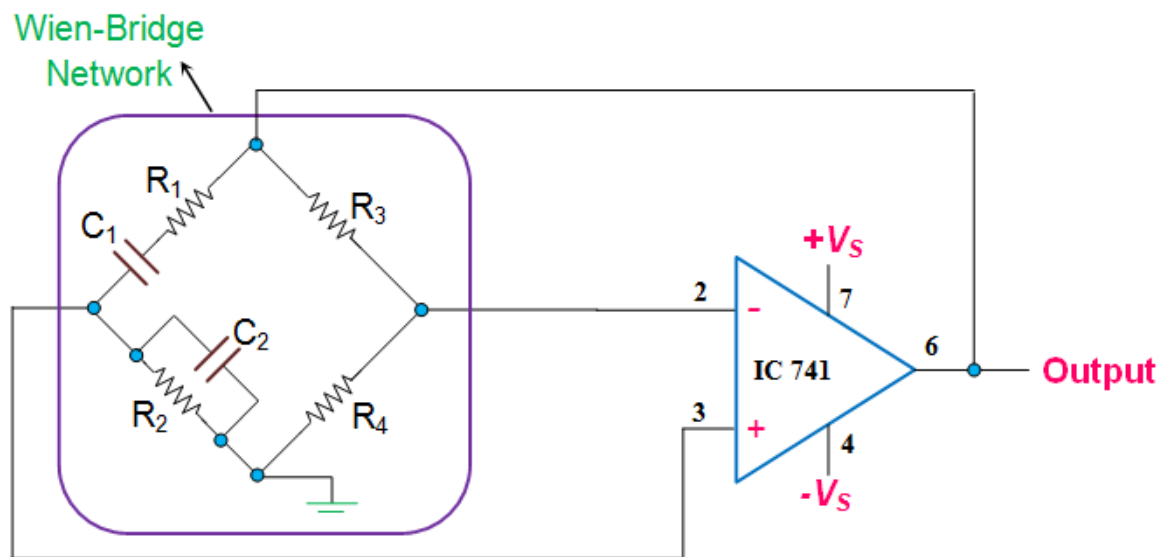
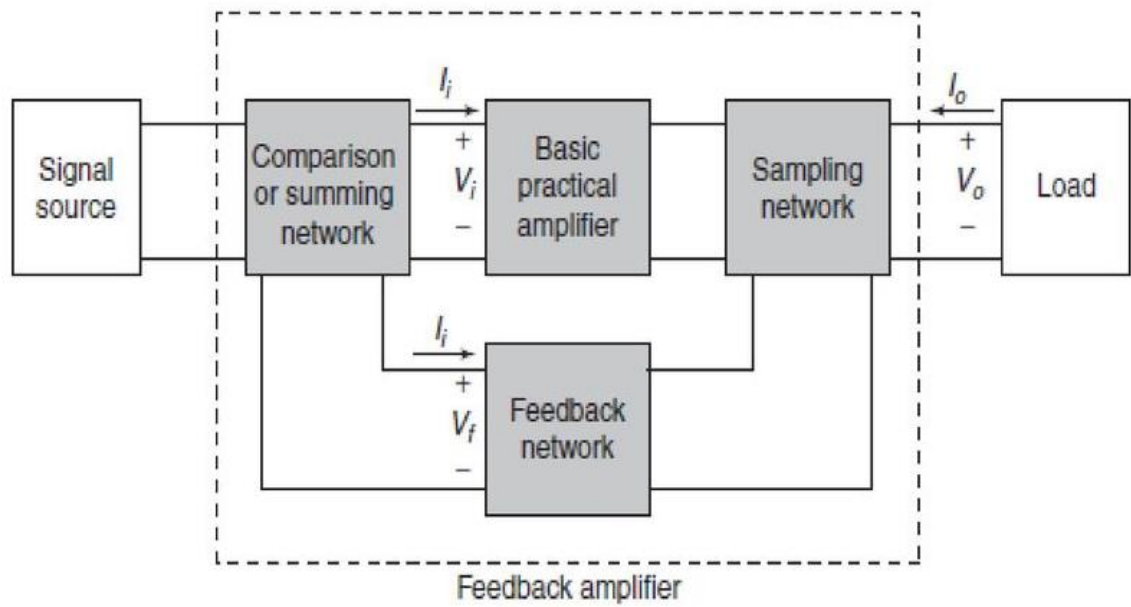


Figure 3 Wien-Bridge Oscillator Using an Op-Amp

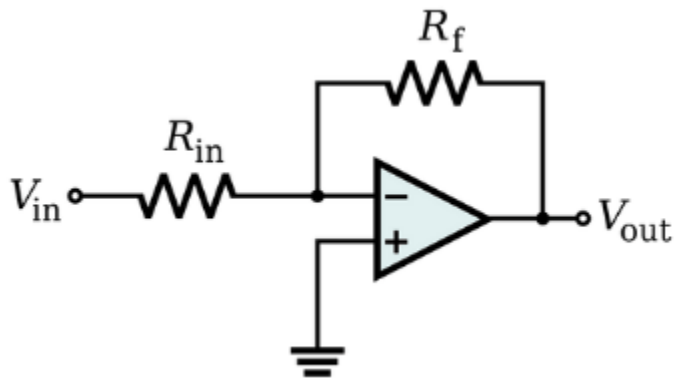
6. BASIC CONCEPTS OF FEEDBACK AMPLIFIER:



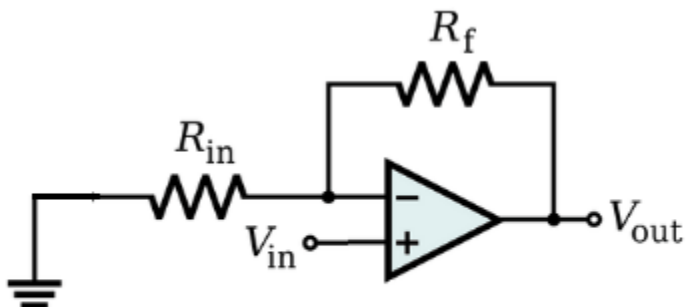
Block diagram of a basic feedback amplifier

UNIT – IV

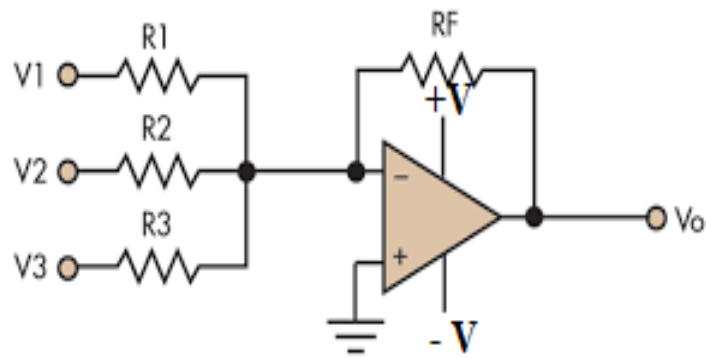
1. Inverting Amplifier:



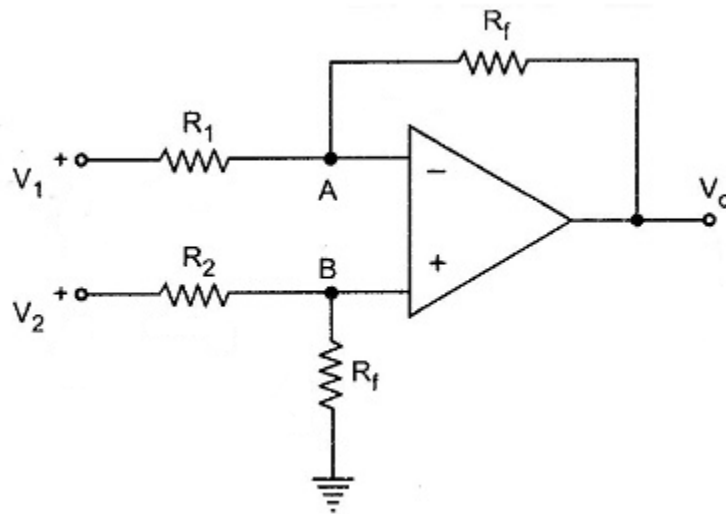
2. Non – Inverting Amplifier:



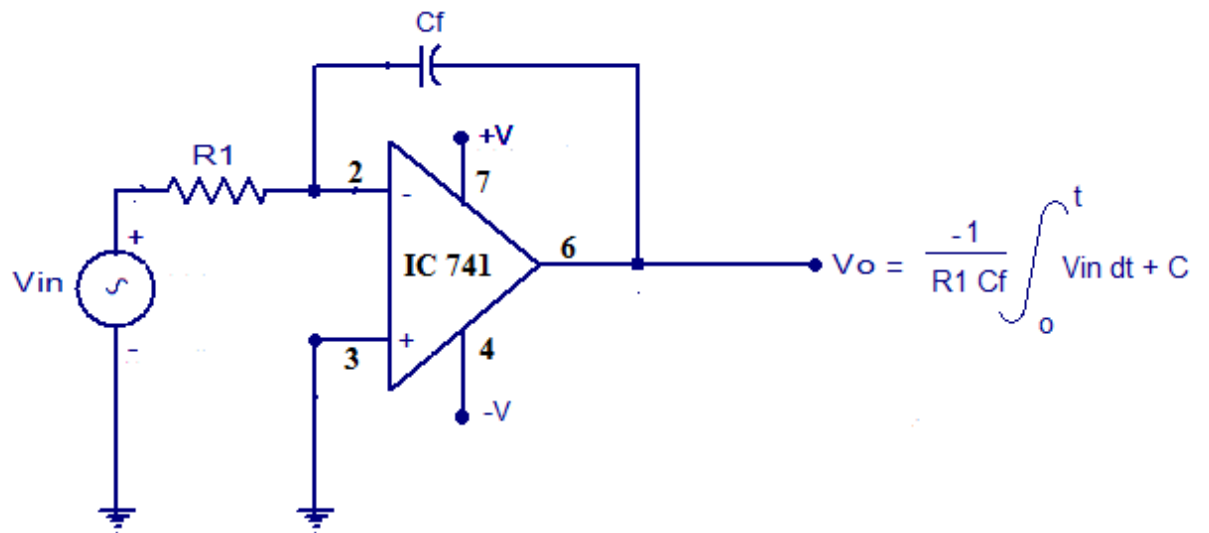
3. Summer:



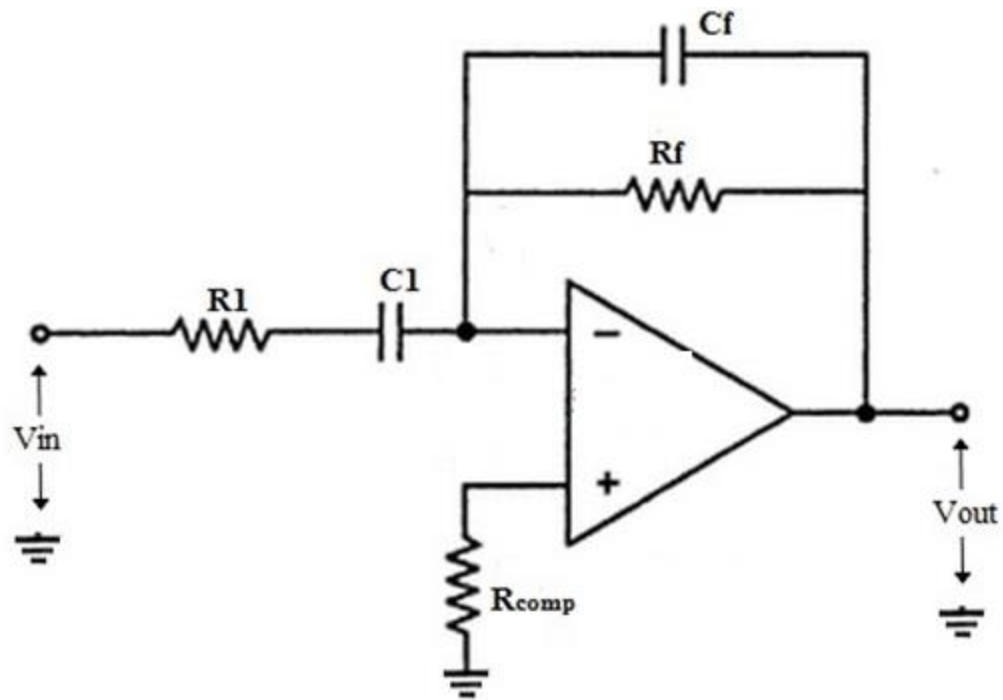
4. Subtractor:



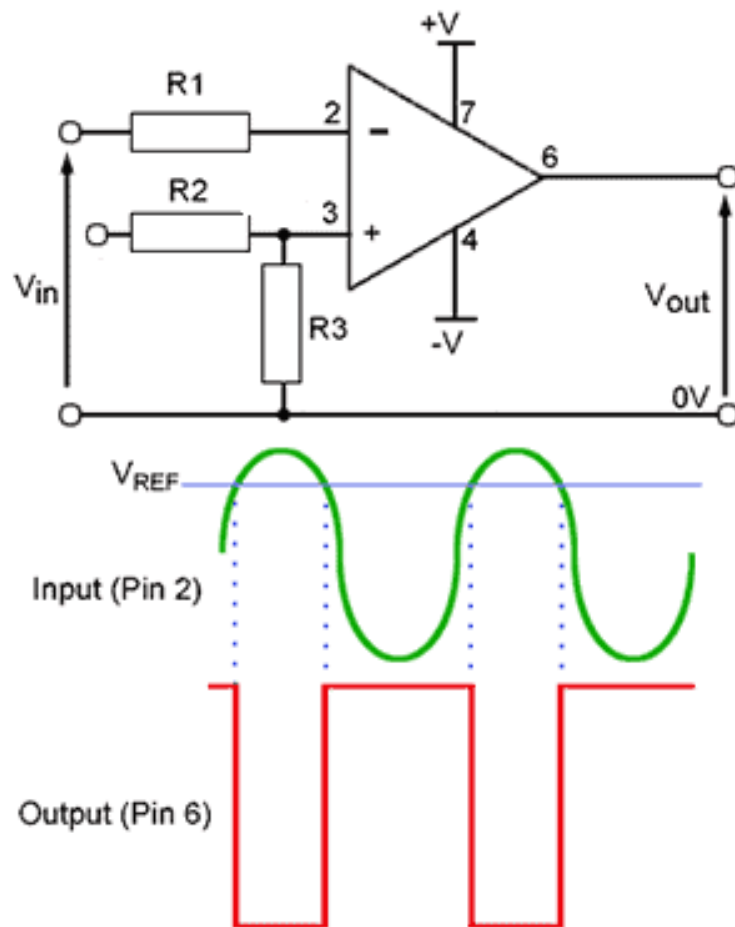
5. Integrator:



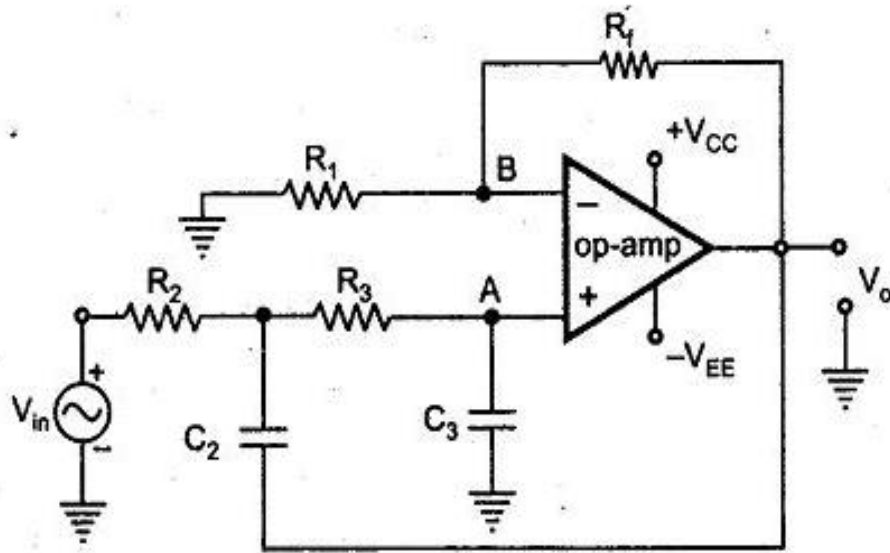
6. Differentiator:



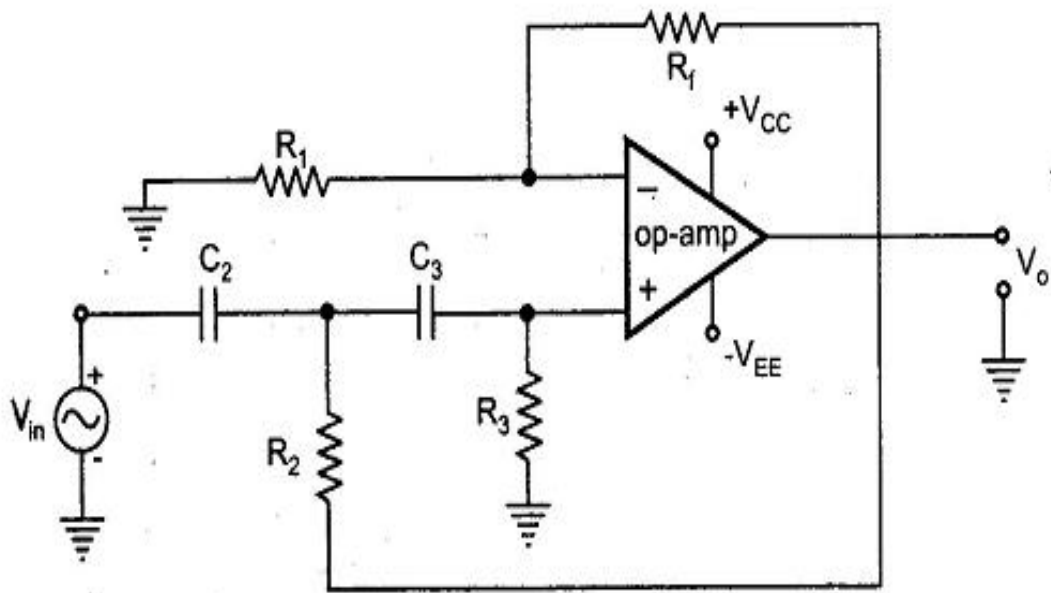
7. Comparator:



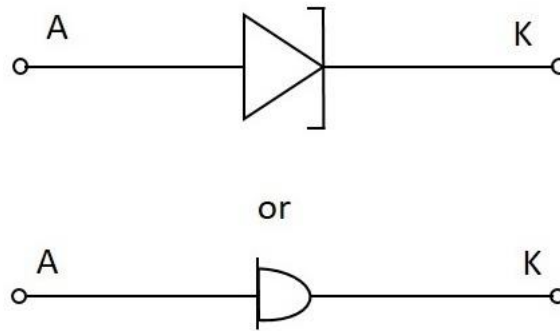
8. Second order Low Pass Filter:



9. Second order High Pass Filter:

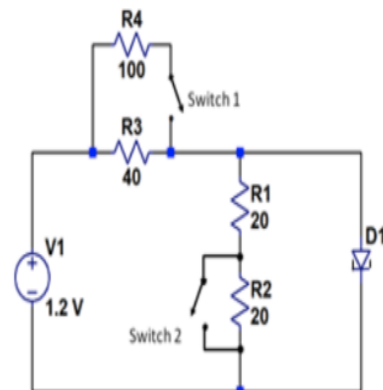
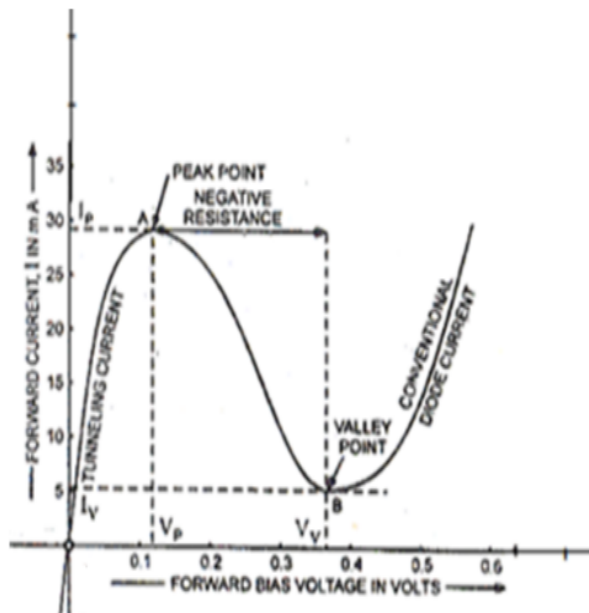


1. TUNNEL DIODE:



Symbol of Tunnel diode

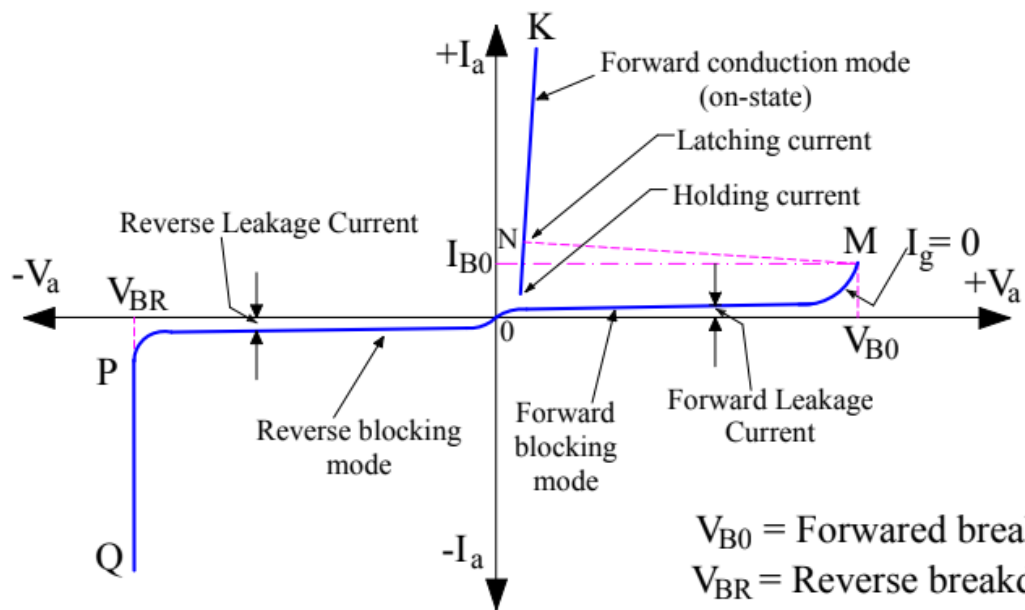
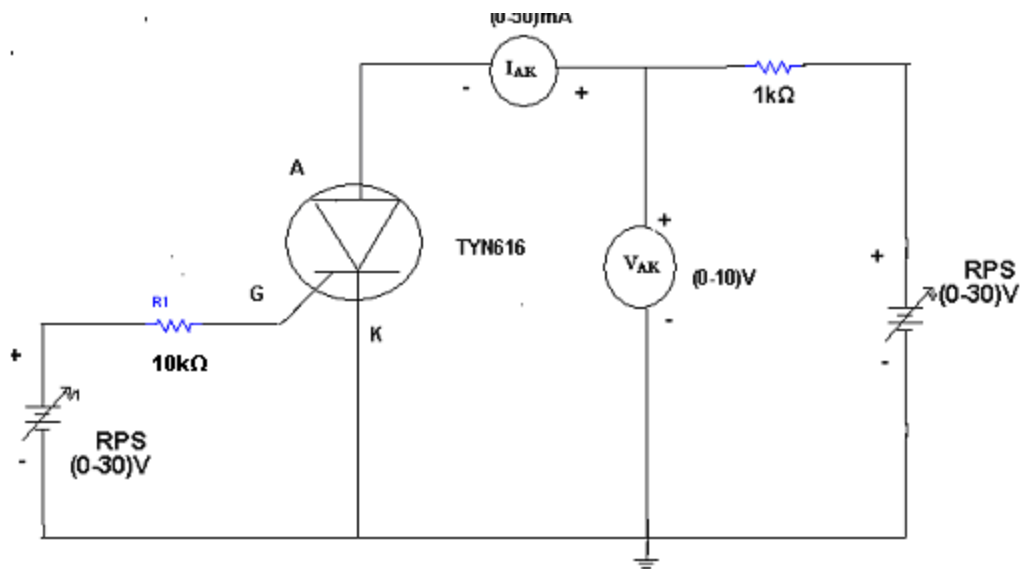
A tunnel diode has the i - v characteristic shown below. The diode is connected to the circuit shown.



###

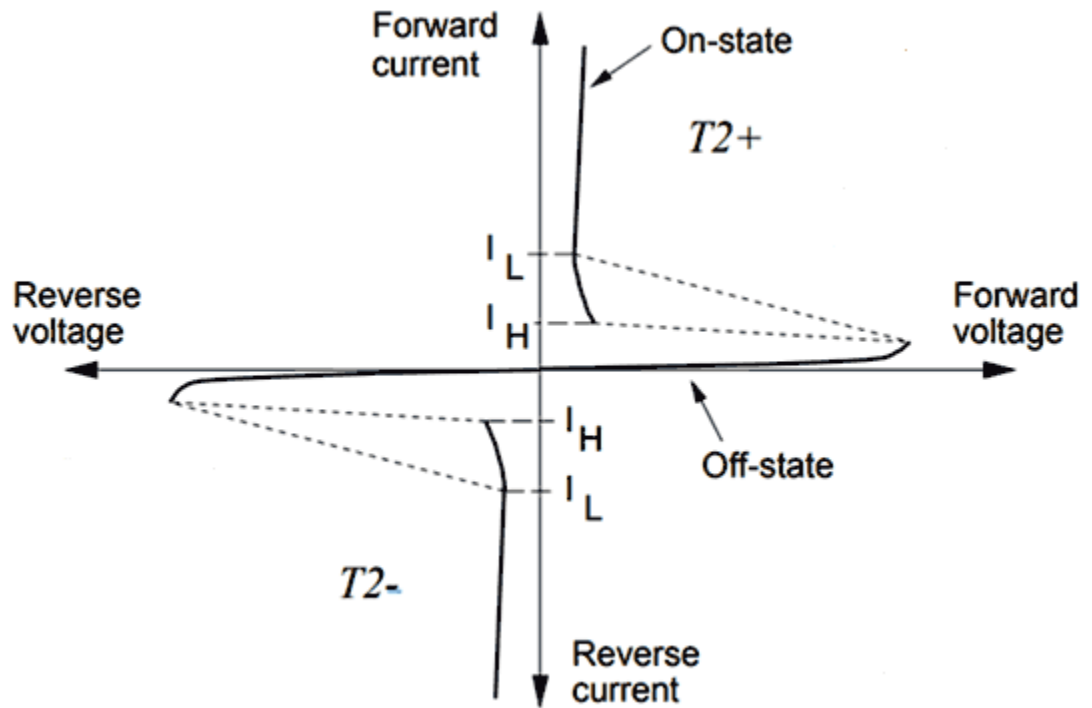
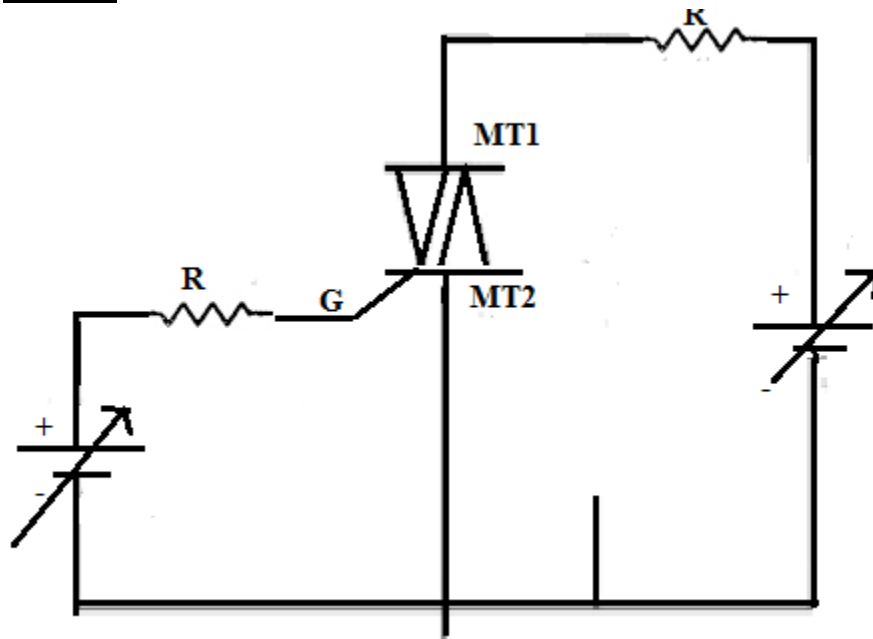
- With both switches *open*, what are the two possible stable operating points for the diode?
- What is the unstable operating point in the negative resistance region?
- The stable operating point with the higher voltage can be forced by momentarily pushing switch #1. What is the operating point of the diode with switch #1 closed?
- The operating point with the lower voltage can be forced by momentarily pushing switch #2. What is the operating point of the diode with switch #2 closed?
- Can you speculate as to why the center operating point is unstable, such that, with both switches open it will quickly revert to one of the two stable points? (Hint: Think about what happens if noise in the system momentarily causes the diode voltage to go up or down by a small, incremental amount.)

2. SILICON CONTROL RECTIFIER(SCR):

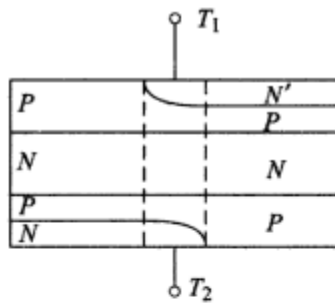
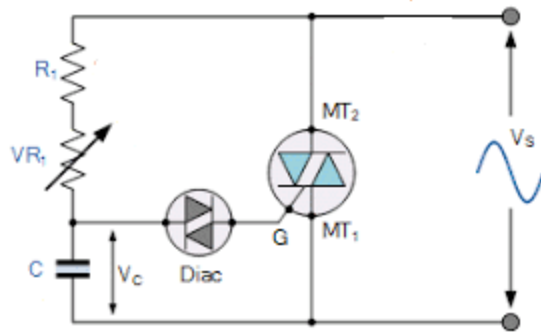


V_{B0} = Forward breakover voltage
 V_{BR} = Reverse breakdown voltage
 I_g = Gate current

3. TRAIC:



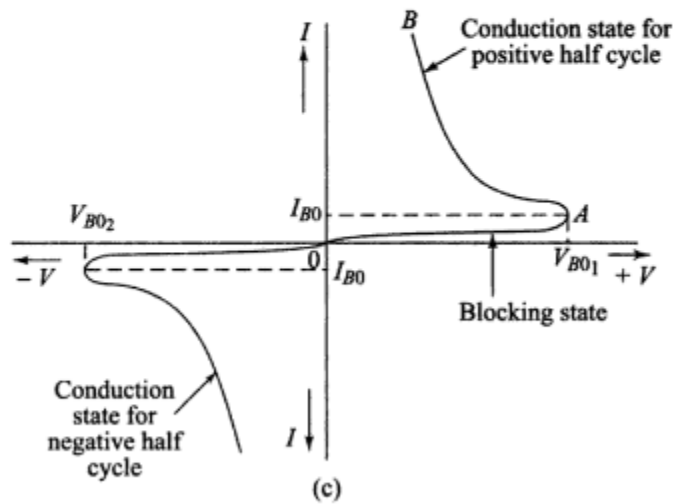
4. DIAC:



(a) Construction



(b) Symbol

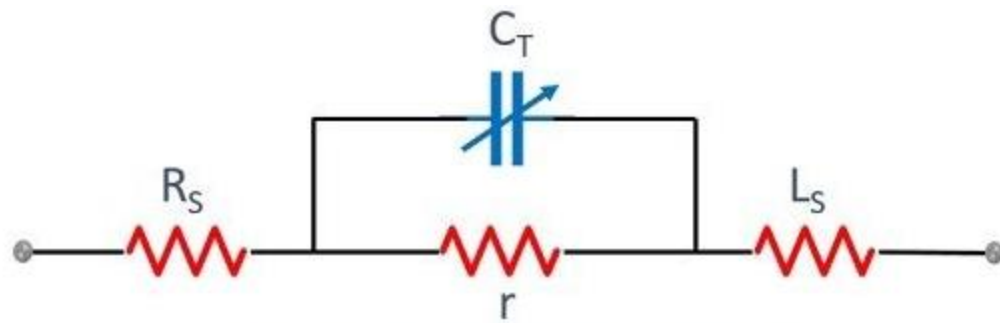


Schematic construction, circuit symbol and V-I characteristics of a diac

5. VARACTOR DIODE:

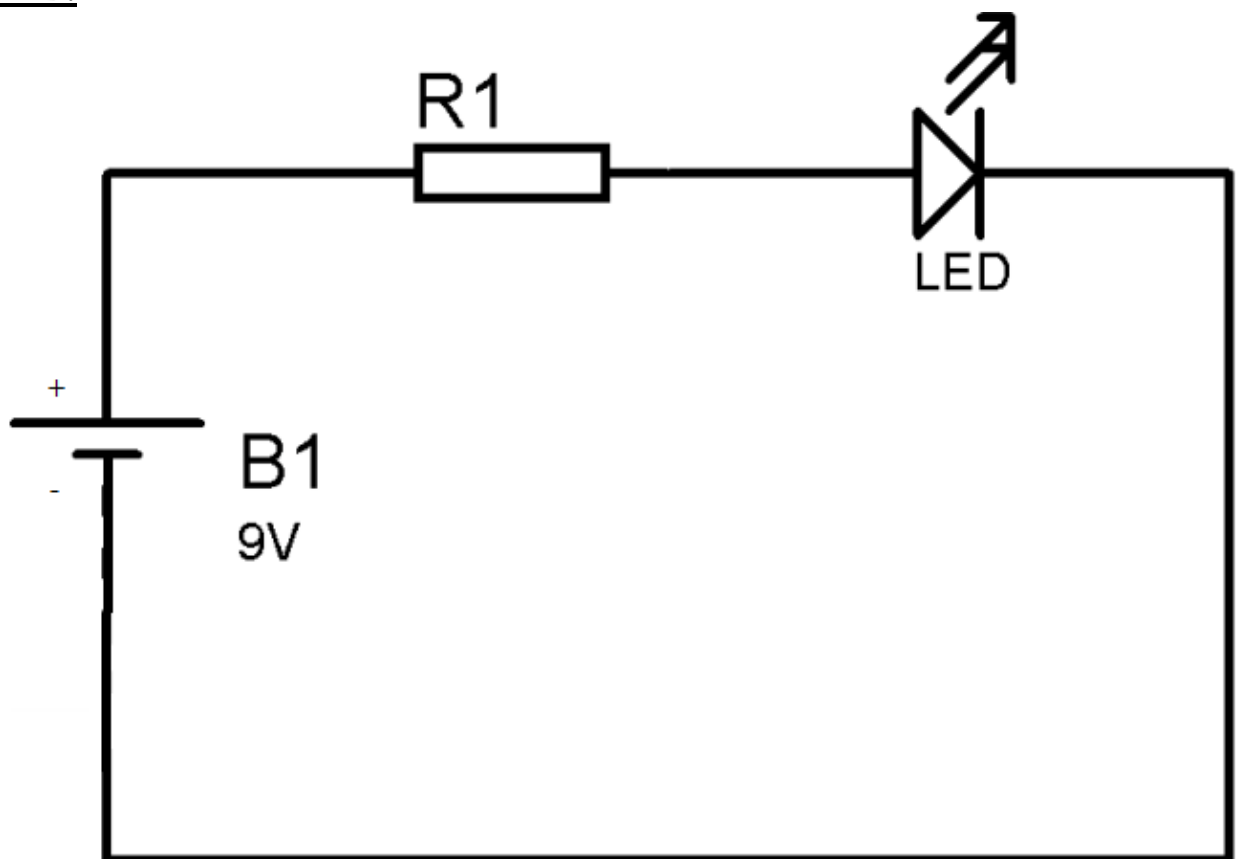
SCHEMATIC SYMBOL



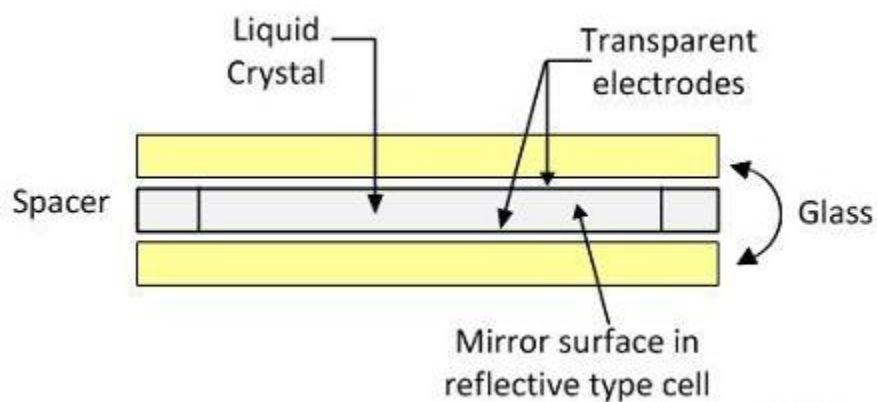


Equivalent circuit of varactor diode in reverse biased condition

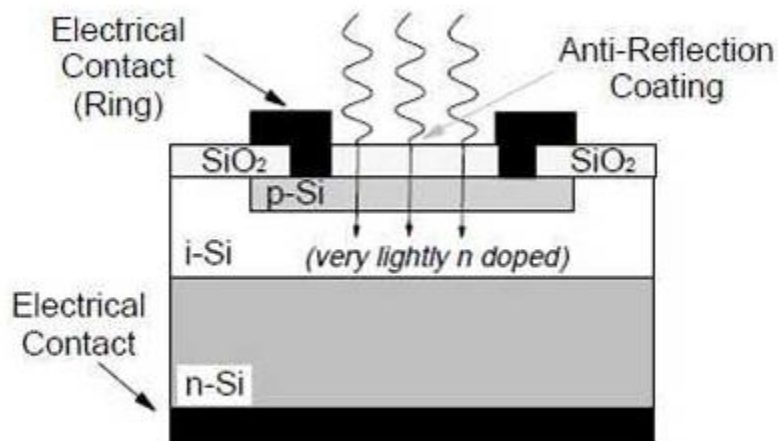
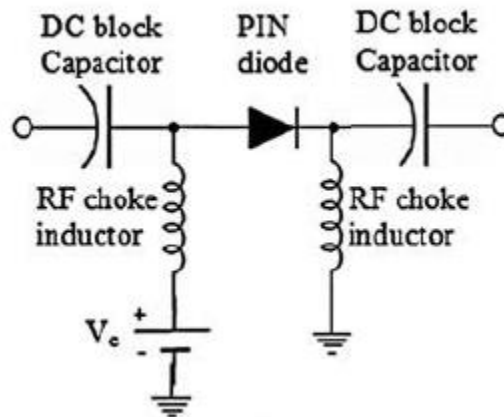
6. LED:



7. LCD:

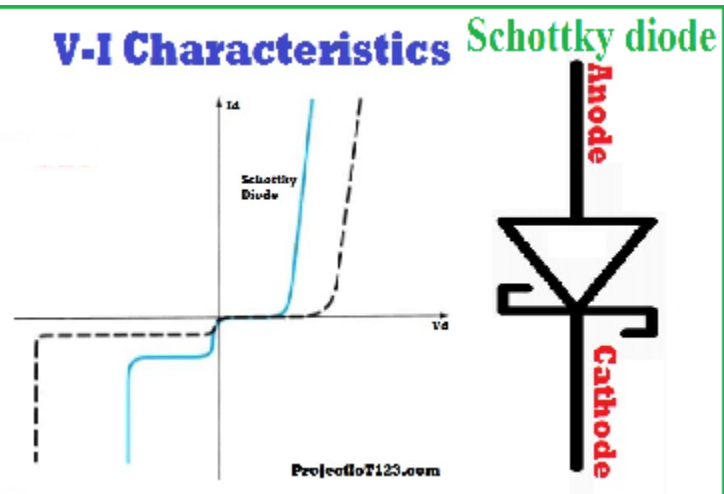


8. PIN DIODE:

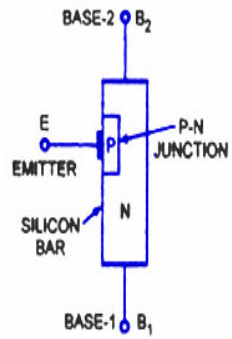


P-I-N Diode Schematic

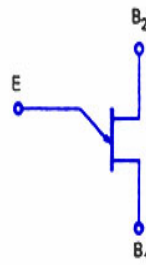
9. SCHOTTKY DIODE:



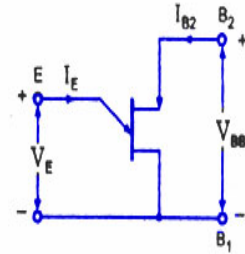
10. UJT:



(a) Basic Structure



(b) Schematic Symbol
Unijunction Transistor



(c) Basic Arrangement